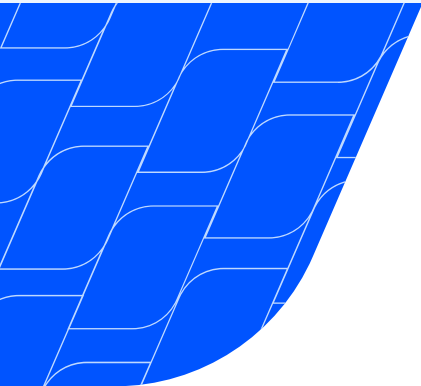




Bank Customer Churn

Analysis & Retention Strategy

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2026



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Executive Summary

Project Context: Customer retention is highly cost-effective compared to new customer acquisition. This project analyzes a dataset of 10,000 bank customers to identify key drivers of churn and develop data-driven strategies to mitigate the attrition of high-net-worth accounts.

Top 3 Business Insights:

- **Churn portrait:** Churn hits **82% - 100%** for customers holding 3-4 products(280 customers lost).
- **High-Value Attrition:** The 46-60 age group holds the highest avg. balance (**\$81.5K**) but suffers a severe **52%** churn rate.
- **Engagement > Products:** "Active Members" with a credit card show the lowest churn (**13.4%**).

Strategic Recommendations:

- **Halt Aggressive Cross-Selling:** Stop pushing 3+ products. Focus on "**Quality 1+1 bundling**".
- **Drive Active Engagement:** Reallocate budget to VIP loyalty programs for the **46-60** high-net-worth demographic.



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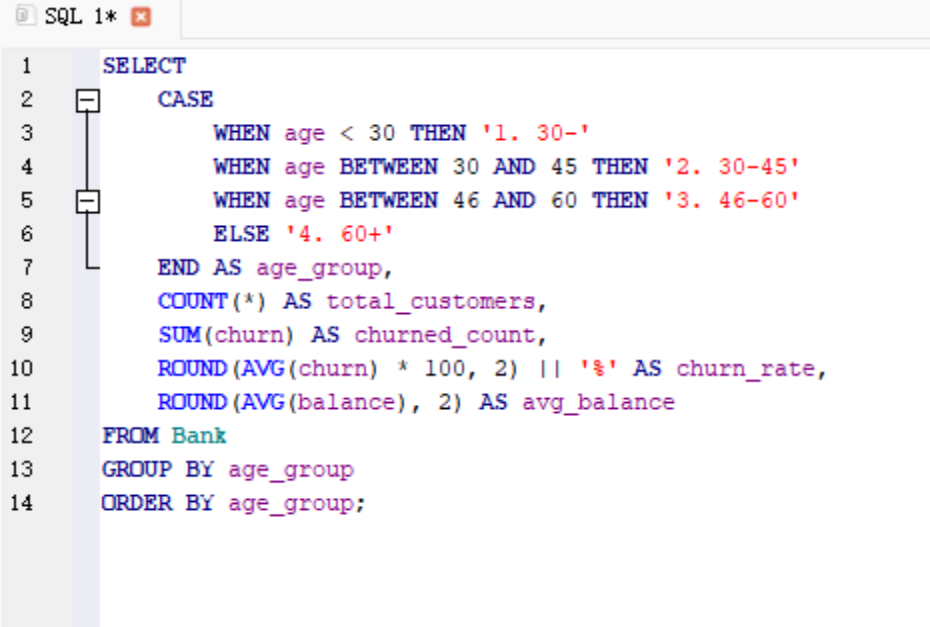
Data Engineering & SQL Logic

Data Processing Approach:

- Structured SQL queries to segment customers and aggregate metrics.

SQL Applied:

- CASE WHEN: Segmented continuous age data into distinct age cohorts (e.g., 30-, 46-60).
- GROUP BY & AVG(): Calculated churn rates across multidimensional segments.



```
1 SELECT
2   CASE
3     WHEN age < 30 THEN '1. 30-'
4     WHEN age BETWEEN 30 AND 45 THEN '2. 30-45'
5     WHEN age BETWEEN 46 AND 60 THEN '3. 46-60'
6     ELSE '4. 60+'
7   END AS age_group,
8   COUNT(*) AS total_customers,
9   SUM(churn) AS churned_count,
10  ROUND(AVG(churn) * 100, 2) || '%' AS churn_rate,
11  ROUND(AVG(balance), 2) AS avg_balance
12 FROM Bank
13 GROUP BY age_group
14 ORDER BY age_group;
```

	age_group	total_customers	churned_count	churn_rate	avg_balance
1	1. 30-	1641	124	7.56%	73698.72
2	2. 30-45	6248	956	15.3%	75951.26
3	3. 46-60	1647	842	51.12%	81500.48
4	4. 60+	464	115	24.78%	75742.6

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Interactive Dashboard

Visualizing the Churn Landscape



■ Financial Risk Quantification:

The dashboard explicitly calculates the potential financial loss from customers holding 4 products (**\$5.62M**), bridging the gap between raw data and business impact.

■ Multidimensional Breakdown:

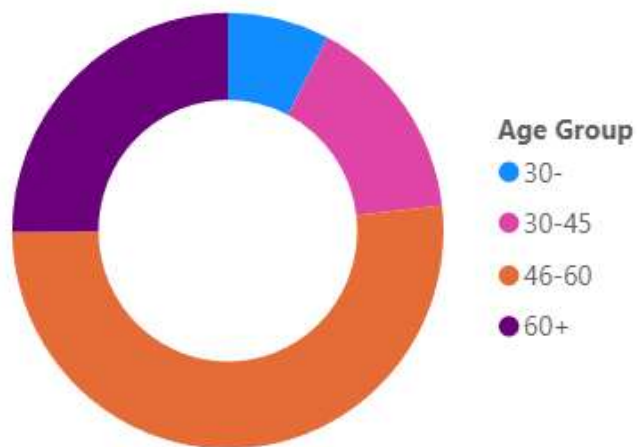
Incorporates pie charts for demographic distribution and clustered column charts to visualize the correlation between product counts, active status, and retention.

Credit Card vs. Active Member

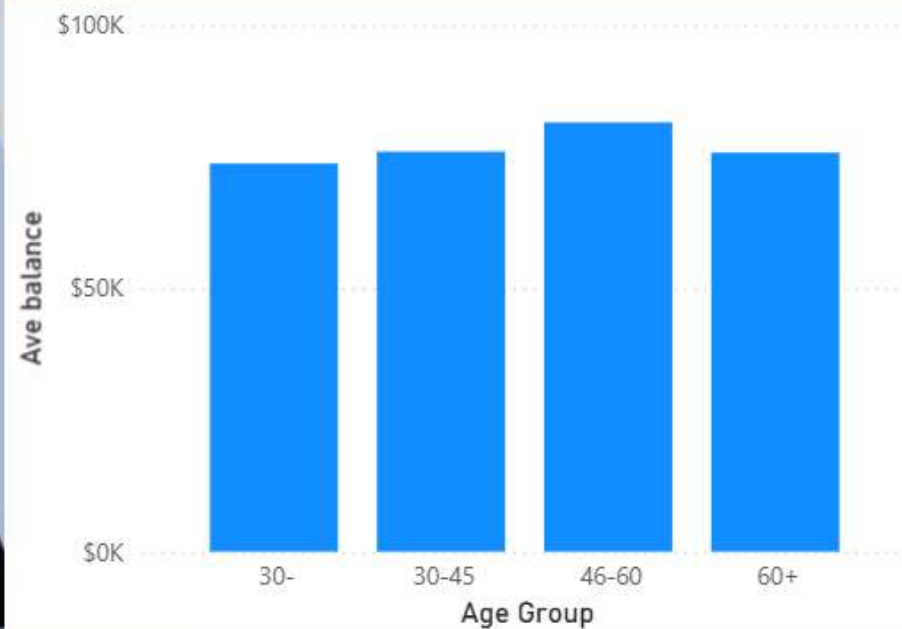
Active Member	0	1
0	25.70%	27.32%
1	16.39%	13.36%

Credit card

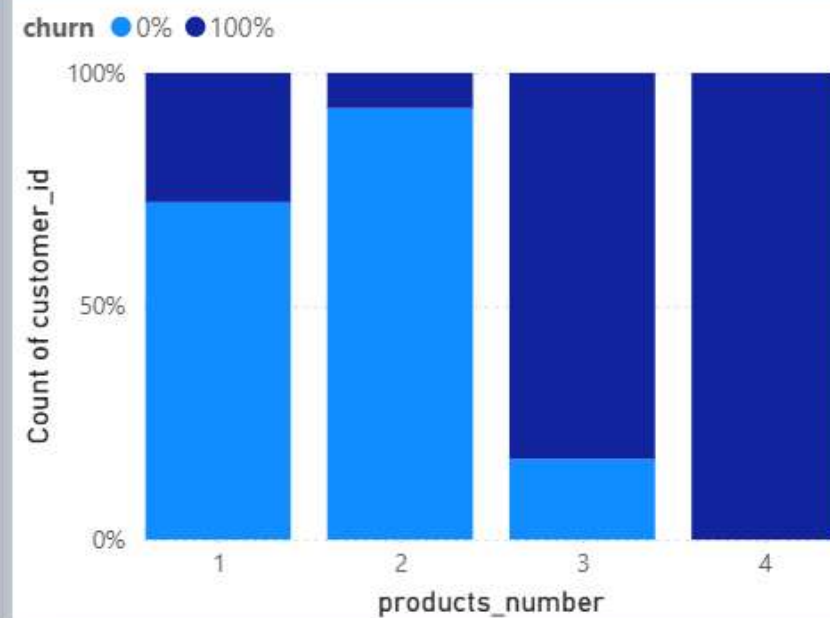
Churn Distribution Across Age Groups



Ave Balance by Age Group



Customer Churn Rate



Losses from customer churn for 4 products

5.62M

/04

Deep Dive & Actionable Strategy

The Product-Churn Anomaly

- **Data:** Churn spikes from **7.6%** (2 products) to **100%** (4 products).
- **Hypothesis:** Aggressive cross-selling over-saturates users, causing dissatisfaction and forced attrition.
- **Action:** Implement a strict "**Marketing Cap**" (Max 2 products). Run A/B tests to discover the golden "1+1" combination and conduct root-cause surveys on product friction.

The 46-60 High-Net-Worth Crisis

- **Data:** Highest churn (**51.1%**) combined with highest avg. balance (**\$81.5K**).
- **Hypothesis:** One-size-fits-all marketing fails to retain affluent demographics who expect tailored relationship management.
- **Action:** Assign dedicated Relationship Managers (RM) and host exclusive offline events to drive "Active Membership".

